

fies the particular key used to encrypt a message out of the set of available keys (either session or group), and the particular encryption/message integrity algorithm to use for the message. The Session ID field is always present. For details on derivation of this field, see respective sections on [unicast](#) and [group](#) session ID derivation.

4.4.1.3. Security Flags (8 bits)

An unsigned integer bit field containing the following subfields:

Table 11. Security Flags field definition

bit 7	6	5	4	3	2	1	0
P	C	MX	Reserved			Session Type	

NOTE

All unused bits in the Security Flags field are reserved and SHALL be set to zero on transmission and SHALL be silently ignored on reception.

P Flag (1 bit, position 7)

The Privacy (P) flag is a single bit field which, when set, SHALL identify that the message is encoded with privacy enhancements as specified in [Section 4.9.3, “Privacy Processing of Outgoing Messages”](#).

C Flag (1 bit, position 6)

The Control message (C) flag is a single bit field which, when set, SHALL identify that the message is a control message, such as for the [Message Counter Synchronization Protocol](#), and uses the control message counter for the nonce field as specified in [Section 4.8.1.1, “Nonce”](#).

MX Flag (1 bit, position 5)

The Message Extensions (MX) flag is a single bit field which, when set, SHALL indicate that the [Message Extensions](#) portion of the message is present and has non-zero length. Version 1.0 Nodes SHALL set this flag to zero.

Session Type (2 bit, position 0-1)

An unsigned integer specifying the type of session associated with the message. The following values are defined:

- 0 — Unicast Session
- 1 — Group Session
- 2-3 — Reserved for future use

Messages with Session Type set to reserved values are not valid and SHALL be dropped without sending a message-layer acknowledgement.

The Session Type defines how the Session ID is to be interpreted.

The *Unsecured Session* SHALL be indicated when both Session Type and Session ID are set to 0. The